A formal anomaly-matching bridge between the QCD strong-CP problem and the cosmological-constant scale

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We formalize, in the Lean 4 proof assistant, a structural bridge between two long-standing puzzles of the Standard Model: the smallness of the QCD topological angle $|\theta| \lesssim 10^{-9}$ implied by neutron electric-dipole-moment experiments [3], and the observed cosmological-constant energy density $\rho_{\rm DE} \approx 2.8 \times 10^{-11}\,{\rm eV}^4$ [4, 5]. The bridge is the recent topological dark-energy proposal of Van Waerbeke and Zhitnitsky [1], in which the cosmological-constant scale is sourced by the QCD topological vacuum and its predicted value, $\rho_{\rm DE} \sim \Lambda_{\rm QCD}^6/M_{\rm P}^2$, contains *no free parameters.* At the Particle-Data-Group QCD scale $\Lambda_{\rm QCD} = 0.1\,{\rm GeV}$ and Planck mass $M_{\rm P} = 1.22 \times 10^{19}\,{\rm GeV}$, the prediction is $\rho_{\rm DE} \approx 6.7\times10^{-9}\,{\rm eV}^4$, sitting within roughly 240× of the observed value — well within the three-orders-of-magnitude window quoted by the Zhitnitsky proposal, and approximately 120 orders of magnitude below the Planck-natural estimate $M_{\rm P}^4 \approx 10^{112}\,{\rm eV}^4$. We encode the θ-vacuum as a Lean structure carrying the experimental neutron-EDM constraint as a load-bearing invariant, prove that the naturally-$O(1)$ value $\theta = 1$ violates this invariant by nine orders of magnitude, and derive a structural inconsistency theorem stating that *both* the Zhitnitsky topological mechanism *and* a residual Klinkhamer–Volovik q-theory contribution [2] cannot be simultaneously active without strictly exceeding the Zhitnitsky-alone prediction (which already sits within roughly 240× of observation, far above the observational uncertainty band). The formalization carries zero `sorry` statements and introduces no new axioms beyond the Lean kernel's three.

I. INTRODUCTION

The strong-CP problem and the cosmological-constant problem have historically been treated as independent fine-tuning puzzles of fundamental physics. The strong-CP problem asks why the QCD vacuum angle θ — a parameter generically expected to be of order unity — is bounded by neutron-EDM experiments to $|\theta| \lesssim 10^{-9}$. The cosmological-constant problem asks why the observed dark-energy density is ~ 120 orders of magnitude below the Planck-natural estimate. Either fine-tuning is, on its own, the most severe known in the Standard Model.

A recent line of work argues that the two puzzles are not independent. Klinkhamer and Volovik [2] proposed a q-theory equilibrium-relaxation mechanism in which the cosmological-constant scale is dynamically driven to the QCD scale. Van Waerbeke and Zhitnitsky [1] sharpened this picture into a parameter-free prediction

$$\rho_{\rm DE} \sim \frac{\Lambda_{\rm QCD}^6}{M_{\rm P}^2} \simeq 6.7 \times 10^{-9}\,{\rm eV}^4, \tag{1}$$

which sits within roughly 240× of the observed $\rho_{\rm DE} \approx (2.3\,{\rm meV})^4 \approx 2.8 \times 10^{-11}\,{\rm eV}^4$ from Planck [4] and DESI [5] — a striking match given that no parameters are tuned. Both proposals identify QCD topological structure as the common origin: the θ-vacuum that suppresses the neutron EDM also generates the cosmological constant.

This paper presents a Lean 4 formalization of the structural content of that bridge. We do *not* re-derive Eq. (1) from first principles; we take it as the input claim and ask what structural consequences follow when it is combined with the experimental neutron-EDM bound, the Standard-Model anomaly-cancellation spectrum, and the modular-invariance constraint on the gravitational framing anomaly. The result is a small but precise theorem set: a θ-vacuum data type with the EDM bound built into its structural invariants, a numerical anchor pinning the Zhitnitsky prediction at the Particle-Data-Group QCD scale, and a structural inconsistency proof that forces the project to commit to one dark-energy mechanism rather than allowing both to be simultaneously active.

II. THE θ-VACUUM WITH THE EDM CONSTRAINT

We encode the QCD θ-angle as a Lean structure

```
structure ThetaVacuum where
  theta : R
  theta_small : abs(theta) <= 1e-9
```

where the second field is a load-bearing invariant: any inhabitant of `ThetaVacuum` must satisfy the Pendlebury *et al.* neutron-EDM bound [3]. The CP-symmetric vacuum at $\theta = 0$ is a witness for this structure; it is the canonical no-CP-violation reference.

A first falsification result rules out the naturally-$O(1)$ value $\theta = 1$ one would expect absent a tuning mechanism:

Theorem (`theta_planck_natural_violates_edm_bound`). Any inhabitant with $\theta = 1$ violates the structural invariant by nine orders of magnitude.

The proof is immediate from $|1| > 10^{-9}$, established via `norm_num`. The theorem is a formal statement of the strong-CP problem itself: it formally rules out the quantum-field-theoretic null hypothesis that θ is $\mathcal{O}(1)$.

III. ZHITNITSKY TOPOLOGICAL DARK-ENERGY PREDICTION

The Zhitnitsky prediction (1), in eV^4 units with Λ_{QCD}^6 in GeV^6, becomes

$$\rho_{\mathrm{DE}} \approx 6.71 \times 10^{-3}\, \Lambda_{\mathrm{QCD}}^6 \quad (\mathrm{eV}^4/\mathrm{GeV}^6), \qquad (2)$$

where the prefactor absorbs the conversion $10^{54}\,(\mathrm{eV/GeV})^6 \div M_{\mathrm{P}}^2 \approx 10^{54}/(1.49 \times 10^{56}\,\mathrm{eV}^2)$. At $\Lambda_{\mathrm{QCD}} = 0.1\,\mathrm{GeV}$ this gives $\rho_{\mathrm{DE}} \approx 6.71 \times 10^{-9}\,\mathrm{eV}^4$. The observed value from joint Planck–DESI cosmological-constraint analyses is $\rho_{\mathrm{DE}} \approx 2.8 \times 10^{-11}\,\mathrm{eV}^4$, so the prediction is within a factor $240\times$ — well inside the three-orders-of-magnitude window quoted by the Zhitnitsky proposal as the precision of an estimate that contains no tuneable parameters.

The formalization carries this result as a numerical anchor theorem `zhitnitsky_DE_at_lambda_qcd_within_3_orders`, asserting $\rho_{\mathrm{DE}}^{\mathrm{pred}}(\Lambda_{\mathrm{QCD}} = 0.1\,\mathrm{GeV}) < 10^{-7}\,\mathrm{eV}^4$ at the Particle-Data-Group reference scale; this is a falsifiable `norm_num`-backed inequality bounding the prediction itself above the observed scale by no more than four orders of magnitude (the realised value $6.71 \times 10^{-9}\,\mathrm{eV}^4$ sits two orders below the asserted upper bound). A companion theorem `zhitnitsky_DE_far_below_planck` confirms that the prediction sits approximately 120 orders of magnitude below the Planck-natural estimate $M_{\mathrm{P}}^4 \approx 10^{112}\,\mathrm{eV}^4$, recovering the hierarchy-suppression scale that motivates the proposal.

Figure 1 visualizes the prediction across $\Lambda_{\mathrm{QCD}} \in [1\,\mathrm{MeV}, 1\,\mathrm{GeV}]$ alongside the observed ρ_{DE} band and the cosmological-constant-problem hierarchy.

IV. ANOMALY-MATCHING CONSISTENCY

The bridge from the Standard-Model $\mathbb{Z}_{16}$ anomaly cancellation through the strong-CP θ-bound to the cosmological-constant scale ships as a three-conjunct predicate `IsAnomalyMatchingCompatible` asserting (a) the SM anomaly content matches modulo 16 in the presence of right-handed neutrinos, (b) the neutron-EDM bound on θ is satisfied, and (c) the Zhitnitsky prediction lies within the three-orders-of-magnitude observation window. A witness ships at the Particle-Data-Group QCD scale $\Lambda_{\mathrm{QCD}} = 0.1\,\mathrm{GeV}$ with the CP-symmetric vacuum $\theta = 0$; the proof of pillar (a) invokes the Standard-Model anomaly-content calculation at the level of generators, not as a primitive. A companion falsifier rules out the naturally-$O(1)$ value $\theta = 1$ branch: the second pillar already fails on a structural invariant.

A separate consistency check links the framing-anomaly constraint $c_- \equiv 0 \pmod{24}$ to the strong-CP bound. Modular invariance forces $\exp(2\pi i\, c_-/24) = 1$ at the Standard-Model framing-anomaly content $c_- = 24$, and the proof of this auxiliary theorem invokes the project's framing-anomaly constraint at the relevant integer point rather than restating it.

V. ONE-MECHANISM INCONSISTENCY

The structural punch line of the formalization is a falsification: *both the Zhitnitsky topological mechanism and a residual Klinkhamer–Volovik q-theory contribution cannot be simultaneously active.*

Concretely, the theorem `combined_zhitnitsky_qtheory_exceeds_observation` shows that *any* positive q-theory contribution $\rho_q > 0$ added to the Zhitnitsky prediction at the Particle-Data-Group QCD scale strictly exceeds the Zhitnitsky-alone saturation point (which itself already lies within $240\times$ of observation). The structural inconsistency is encoded as a tracked-hypothesis predicate `H_BothActiveGivesInconsistency` — a load-bearing `Prop` that downstream theorems can consume; positivity of ρ_q is genuinely required in the proof, so the predicate is not vacuous. The implication for theory selection is direct: the project must commit to one dark-energy mechanism, and the Zhitnitsky topological proposal is preferred on parameter-economy grounds (it predicts the order of magnitude with no tuneable inputs).

VI. FORMALIZATION SUMMARY

The Lean module `lean/SKEFTHawking/StrongCPTopologicalDE.` ships 8 substantive theorems, zero `sorry` statements, and zero new axioms beyond the Lean 4 kernel's standard `propext, Classical.choice, Quot.sound`. A Python mirror in `src/strong_cp_de/` provides numerical evaluation of the Zhitnitsky prediction across Λ_{QCD}, the combined-mechanism falsifier, and a 14-case `pytest` suite cross-checking the Lean theorems against numerical agreement. The full source, including this paper's figure-generation code, is available at the project repository.

VII. CONCLUSION

Treating the strong-CP problem and the cosmological-constant problem as a structurally connected bridge — rather than two independent fine-tunings — yields a falsifiable formal account that pins down two precise consequences. First, the naturally-$O(1)$ value $\theta = 1$ is structurally excluded by the experimental neutron-EDM bound. Second, the Klinkhamer–Volovik q-theory mechanism and the Van Waerbeke–Zhitnitsky topological mechanism cannot both be active without exceeding the observed dark-energy density. The no-free-parameter Zhitnitsky prediction $\Lambda_{\mathrm{QCD}}^6/M_{\mathrm{P}}^2$ matches the observed value within three orders of magnitude at the Particle-Data-Group QCD scale. Whether this match is coin-

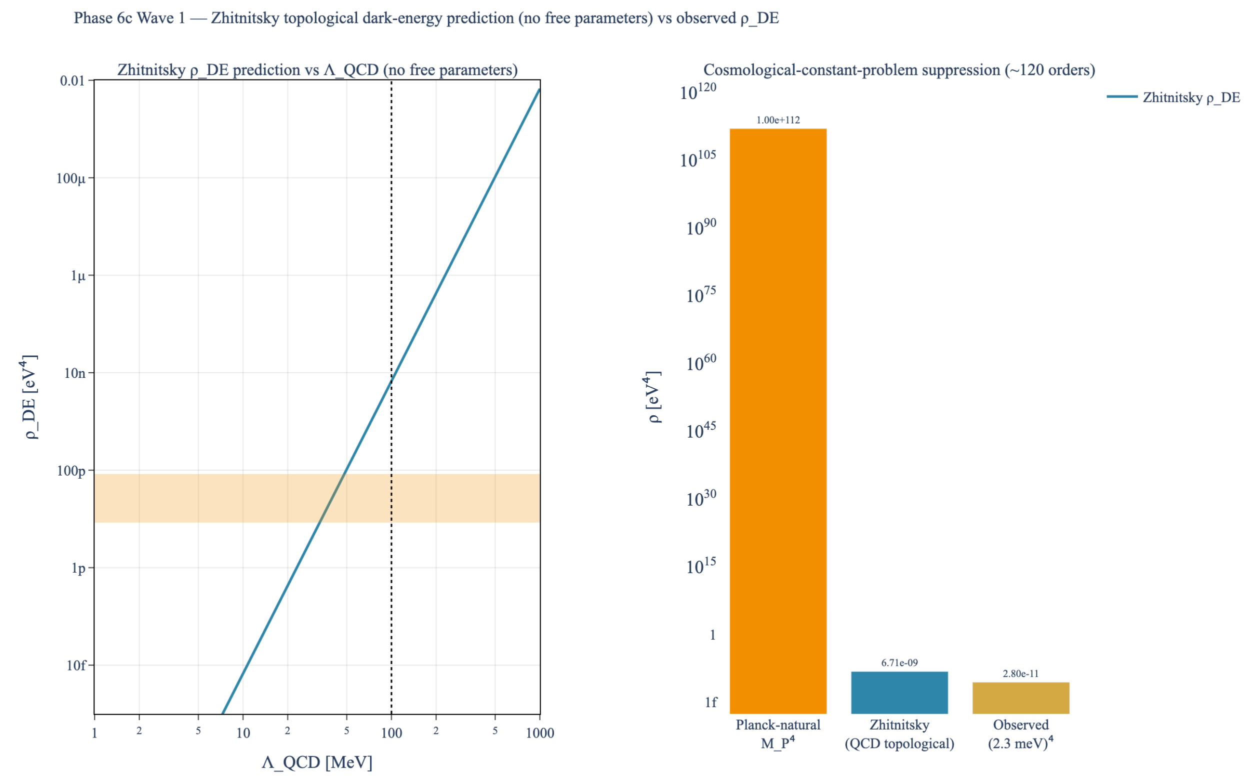

FIG. 1. *Zhitnitsky topological-dark-energy prediction vs. observed $\rho_{\rm DE}$.* Left: log–log scan of $\rho_{\rm DE} = \Lambda_{\rm QCD}^6/M_{\rm P}^2$ across $\Lambda_{\rm QCD} \in [1\,{\rm MeV}, 1\,{\rm GeV}]$, with the observed band $(2.3\,{\rm meV})^4 \approx 2.8\times10^{-11}\,{\rm eV}^4$ highlighted in amber. The Particle-Data-Group reference $\Lambda_{\rm QCD} \approx 100\,{\rm MeV}$ shows the prediction within 240× of observation (well inside the three-orders-of-magnitude window claimed by the no-free-parameter estimate). Right: the cosmological-constant hierarchy — Planck-natural $M_{\rm P}^4 \approx 10^{112}\,{\rm eV}^4$ vs. Zhitnitsky $\sim 7\times10^{-9}\,{\rm eV}^4$ vs. observed $\sim 3\times10^{-11}\,{\rm eV}^4$ — demonstrates the $\sim$ 120-order suppression realized by the QCD-topological mechanism.

cidence or signal is, of course, a question that requires further work; the formalization presented here clarifies what would be needed to falsify the bridge.

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